Historial de pruebas y errores encontrados durante el desarrollo de la actividad Laboratorio 1

Pablo Jiménez Fuentes

BIG DATA APLICADO

Pruebas

Ejecución del docker

```
#13 exporting to image
#13 exporting layers
#13 exporting layers 2.3s done
#13 exporting manifest sha256:045772d84469c64c62305813f6e05aec35df79d83bcf7a71db3bc6a0c1b35952 done
#13 exporting config sha256:65111ffb8bc857fcd993914f72fb12453c45082795c48a4de6b8c4e7d8226135 done
#13 exporting attestation manifest sha256:be923d24bf439f7ea7ddacad8509505f6179d0bade988bdf0edcc7e4a75dcd2e 0.0s done
#13 exporting manifest list sha256:64c9677c1c3ab5965fd75755165ef8ed0354594c61a4f237ecffcc081c931c82 done
#13 naming to docker.io/library/pruebadocker-ingesta:latest done
#13 unpacking to docker.io/library/pruebadocker-ingesta:latest
#13 unpacking to docker.io/library/pruebadocker-ingesta:latest 0.9s done
#13 DONE 3.4s

[+] up 20/20g provenance for metadata file
 ✔Image mysql:8.0                Pulled                                                                  14.2s
 ✔Image pruebadocker-ingesta     Built                                                                   30.6s
 ✔Volume pruebadocker_mysql_data Created                                                                  0.0s
 ✔Container hdfs-namenode        Running                                                                  0.0s
 ✔Container hdfs-datanode        Running                                                                  0.0s
 ✔Container mysql-ventas         Started                                                                  0.4s
 ✔Container ingesta              Started                                                                  0.4s
PS C:\Users\EXPERTO\Desktop\PruebaDocker> docker exec ingesta hdfs dfs -mkdir -p /user/iabd_admin
PS C:\Users\EXPERTO\Desktop\PruebaDocker> docker exec ingesta hdfs dfs -ls /
Found 1 items
drwxr-xr-x   - root supergroup          0 2026-11-03 18:53 /user
PS C:\Users\EXPERTO\Desktop\PruebaDocker> docker exec ingesta sqoop version
Warning: /opt/sqoop/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/sqoop/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
```

Vision de servicios lanzados

```
PS C:\Users\EXPERTO\Desktop\PruebaDocker> docker compose ps
NAME            IMAGE                                          COMMAND                  SERVICE        CREATED
   STATUS                   PORTS
hdfs-datanode   bde2020/hadoop-datanode:2.0.0-hadoop3.2.1-java8   "/entrypoint.sh /run…"   datanode       17 minutes ago
   Up 17 minutes (healthy)   0.0.0.0:9864->9864/tcp, [::]:9864->9864/tcp
hdfs-namenode   bde2020/hadoop-namenode:2.0.0-hadoop3.2.1-java8   "/entrypoint.sh /run…"   namenode       17 minutes ago
   Up 17 minutes (healthy)   0.0.0.0:9000->9000/tcp, [::]:9000->9000/tcp, 0.0.0.0:9870->9870/tcp, [::]:9870->9870/tcp
ingesta         pruebadocker-ingesta                           "/entrypoint.sh tail…"   ingesta        2 minutes ago
   Up 2 minutes
jupyter-lab     continuumio/anaconda3:latest                   "bash -c ' pip insta…"   jupyter        17 minutes ago
   Up 17 minutes             0.0.0.0:8888->8888/tcp, [::]:8888->8888/tcp
kafka           confluentinc/cp-kafka:7.5.0                    "/etc/confluent/dock…"   kafka          17 minutes ago
   Up 17 minutes             0.0.0.0:9092->9092/tcp, [::]:9092->9092/tcp, 0.0.0.0:29092->29092/tcp, [::]:29092->29092/tc
p
mongodb         mongo:7.0                                      "docker-entrypoint.s…"   mongodb        17 minutes ago
   Up 17 minutes             0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp
mysql-ventas    mysql:8.0                                      "docker-entrypoint.s…"   mysql          2 minutes ago
   Up 2 minutes              0.0.0.0:3306->3306/tcp, [::]:3306->3306/tcp
postgres        postgres:16-alpine                             "docker-entrypoint.s…"   postgres       17 minutes ago
   Up 17 minutes             0.0.0.0:5432->5432/tcp, [::]:5432->5432/tcp
spark-master    apache/spark:3.5.1                             "/opt/entrypoint.sh …"   spark-master   17 minutes ago
   Up 17 minutes             0.0.0.0:7077->7077/tcp, [::]:7077->7077/tcp, 0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
spark-worker    apache/spark:3.5.1                             "/opt/entrypoint.sh …"   spark-worker   17 minutes ago
   Up 17 minutes             0.0.0.0:8081->8081/tcp, [::]:8081->8081/tcp
zookeeper       zookeeper:latest                               "/docker-entrypoint.…"   zookeeper      17 minutes ago
```

Entro en mysql

```
PS C:\Users\EXPERTO\Desktop\PruebaDocker> docker exec -it mysql-ventas mysql -u sqoopuser -p'sqooppassword' ventas_db
mysql: [Warning] Using a password on the command line interface can be insecure.
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 13
Server version: 8.0.46 MySQL Community Server - GPL

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owners.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

mysql> CREATE TABLE clientes (
    ->   id_cliente INT PRIMARY KEY,
    ->   nombre VARCHAR(50), ciudad VARCHAR(50), email VARCHAR(80)
    -> );
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE pedidos (
    ->   id_pedido INT PRIMARY KEY,
    ->   id_cliente INT, fecha DATE, producto VARCHAR(50),
    ->   cantidad INT, importe DECIMAL(10,2),
    ->   FOREIGN KEY (id_cliente) REFERENCES clientes(id_cliente)
    -> );
Query OK, 0 rows affected (0.01 sec)

mysql>
```

Creo tablas bd

```
ysql> CREATE TABLE pedidos (
    ->   id_pedido INT PRIMARY KEY,
    ->   id_cliente INT, fecha DATE, producto VARCHAR(50),
    ->   cantidad INT, importe DECIMAL(10,2),
    ->   FOREIGN KEY (id_cliente) REFERENCES clientes(id_cliente)
    -> );
uery OK, 0 rows affected (0.01 sec)

ysql>
ysql> INSERT INTO clientes
    -> WITH RECURSIVE seq AS (SELECT 1 AS n UNION ALL SELECT n+1 FROM seq WHERE n < 100)
    -> SELECT n, CONCAT('Cliente ', n),
    ->        ELT(1 + n % 5, 'Sevilla','Madrid','Barcelona','Valencia','Bilbao'),
    ->        CONCAT('cliente', n, '@ejemplo.com')
    -> FROM seq;
uery OK, 100 rows affected (0.02 sec)
ecords: 100  Duplicates: 0  Warnings: 0

ysql>
ysql> INSERT INTO pedidos
    -> WITH RECURSIVE seq AS (SELECT 1 AS n UNION ALL SELECT n+1 FROM seq WHERE n < 1000)
    -> SELECT n, 1 + FLOOR(RAND()*100),
    ->        DATE_ADD('2025-01-01', INTERVAL FLOOR(RAND()*300) DAY),
    ->        ELT(1 + n % 4, 'Porttil','Ratn','Monitor','Teclado'),
    ->        1 + FLOOR(RAND()*5), ROUND(10 + RAND()*990, 2)
    -> FROM seq;
uery OK, 1000 rows affected (0.01 sec)
```

Lanzo sqoop

```
PS C:\Users\EXPERTO\Desktop\PruebaDocker> docker exec -it ingesta bash
root@ingesta:~# sqoop import \
> --connect jdbc:mysql://mysql:3306/ventas_db \
> --username sqoopuser \
> --password 'sqooppassword' \
> --table pedidos \
> --target-dir /user/iabd_admin/ingesta/ventas/pedidos \
> -m 1
Warning: /opt/sqoop/../hbase does not exist! HBase imports will fail.
Please set $HBASE_HOME to the root of your HBase installation.
Warning: /opt/sqoop/../hcatalog does not exist! HCatalog jobs will fail.
Please set $HCAT_HOME to the root of your HCatalog installation.
Warning: /opt/sqoop/../accumulo does not exist! Accumulo imports will fail.
Please set $ACCUMULO_HOME to the root of your Accumulo installation.
Warning: /opt/sqoop/../zookeeper does not exist! Accumulo imports will fail.
Please set $ZOOKEEPER_HOME to the root of your Zookeeper installation.
/opt/hadoop-3.2.1/libexec/hadoop-functions.sh: line 2401: HADOOP_ORG.APACHE.SQOOP.SQOOP_USER: bad substitution
/opt/hadoop-3.2.1/libexec/hadoop-functions.sh: line 2366: HADOOP_ORG.APACHE.SQOOP.SQOOP_USER: bad substitution
/opt/hadoop-3.2.1/libexec/hadoop-functions.sh: line 2461: HADOOP_ORG.APACHE.SQOOP.SQOOP_OPTS: bad substitution
2026-11-03 19:01:53,170 INFO sqoop.Sqoop: Running Sqoop version: 1.4.7
2026-11-03 19:01:53,230 WARN tool.BaseSqoopTool: Setting your password on the command-line is insecure. Consider using -
P instead.
2026-11-03 19:01:53,287 INFO manager.MySQLManager: Preparing to use a MySQL streaming resultset.
2026-11-03 19:01:53,287 INFO tool.CodeGenTool: Beginning code generation
Tue Nov 03 19:01:53 UTC 2026 WARN: Establishing SSL connection without server's identity verification is not recommended
. According to MySQL 5.5.45+, 5.6.26+ and 5.7.6+ requirements SSL connection must be established by default if explicit
option isn't set. For compliance with existing applications not using SSL the verifyServerCertificate property is set to
```

Relanzo sqoop ya que dio errores por un directorio vacio

```
root@ingesta:~# sqoop import --connect jdbc:mysql://mysql:3306/ventas_db --username sqoopuser --password 'sqooppassword'
  --table pedidos --target-dir /user/iabd_admin/ingesta/ventas/pedidos --bindir /opt/sqoop/lib -m 1
```

Comprobación correcta

```
root@ingesta:~# hdfs dfs -ls /user/iabd_admin/ingesta/ventas
Found 1 items
drwxr-xr-x   - root supergroup          0 2026-11-03 19:10 /user/iabd_admin/ingesta/ventas/pedidos
root@ingesta:~# hdfs dfs -ls /user/iabd_admin/ingesta/ventas/pedidos
Found 2 items
-rw-r--r--   3 root supergroup          0 2026-11-03 19:10 /user/iabd_admin/ingesta/ventas/pedidos/_SUCCESS
-rw-r--r--   3 root supergroup      33978 2026-11-03 19:10 /user/iabd_admin/ingesta/ventas/pedidos/part-m-00000
root@ingesta:~# hdfs dfs -cat /user/iabd_admin/ingesta/ventas/pedidos/part-m-00000 | head
2026-11-03 19:11:45,772 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteH
ostTrusted = false
5,2025-01-30,90,1,791.12,Ratn
1,2025-02-03,50,2,39.61,Monitor
1,2025-06-17,93,3,355.15,Teclado
4,2025-06-28,73,4,167.27,Porttil
1,2025-07-20,43,5,257.46,Ratn
3,2025-08-11,10,6,868.44,Monitor
4,2025-08-25,8,7,182.19,Teclado
5,2025-03-04,75,8,427.69,Porttil
4,2025-02-08,68,9,629.97,Ratn
4,2025-08-14,33,10,627.20,Monitor
root@ingesta:~# hdfs dfs -cat /user/iabd_admin/ingesta/ventas/pedidos/part-m-00000 | wc -l
2026-11-03 19:11:48,341 INFO sasl.SaslDataTransferClient: SASL encryption trust check: localHostTrusted = false, remoteH
ostTrusted = false
1000
root@ingesta:~# echo 'export HADOOP_CLASSPATH=$HADOOP_CLASSPATH:/opt/sqoop/lib' >> ~/.bashrc
root@ingesta:~#
```

Prueba de comunicación entre flume

```
^Croot@ingesta:~# cls
bash: cls: command not found
root@ingesta:~# clear
root@ingesta:~# flume-ng agent \
> --conf /opt/flume/conf \
> --conf-file ~/flume_configs/weblogs.conf \
> --name agent \
> -Xmx512m \
> -Dflume.root.logger=INFO,console
Info: Including Hadoop libraries found via (/opt/hadoop-3.2.1/bi
/opt/hadoop-3.2.1/libexec/hadoop-functions.sh: line 2401: HADOOP
/opt/hadoop-3.2.1/libexec/hadoop-functions.sh: line 2366: HADOOP
/opt/hadoop-3.2.1/libexec/hadoop-functions.sh: line 2461: HADOOP
Info: Including Hive libraries found via () for Hive access
+ exec /usr/lib/jvm/java-8-openjdk-amd64//bin/java -Xmx20m -Xmx5
etc/hadoop:/opt/hadoop-3.2.1/share/hadoop/common/lib/*:/opt/hado
op-3.2.1/share/hadoop/hdfs/lib/*:/opt/hadoop-3.2.1/share/hadoop/
e/hadoop/mapreduce/*:/opt/hadoop-3.2.1/share/hadoop/yarn:/opt/ha
b/*' -Djava.library.path=:/opt/hadoop-3.2.1/lib/native org.apach
conf --name agent
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/flume/lib/log4j-slf4j-imp
SLF4J: Found binding in [jar:file:/opt/hadoop-3.2.1/share/hadoop
s]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4j
```

```
Windows PowerShell
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PS C:\Users\EXPERTO\Desktop\PruebaDocker\ingesta> docker exec -it ingesta bash
root@ingesta:~# echo "192.168.1.1 - GET /index.html" > ~/log_temp.txt
root@ingesta:~# echo "192.168.1.5 - POST /login.php" >> ~/log_temp.txt
root@ingesta:~# mv ~/log_temp.txt /home/iabd_admin/flume_ingest/log_file_01.txt
root@ingesta:~# hdfs dfs -ls /user/iabd_admin/ingesta/weblogs
ls: `/user/iabd_admin/ingesta/weblogs': No such file or directory
root@ingesta:~# ls -la /home/iabd_admin/flume_ingest
total 20
drwxr-xr-x 1 root root 4096 Nov  5 17:53 .
drwxr-xr-x 1 root root 4096 Nov  5 17:53 ..
drwxr-xr-x 2 root root 4096 Nov  3 19:18 .flumespool
-rw-r--r-- 1 root root   60 Nov  5 17:53 log_file_01.txt
root@ingesta:~# jps
617 Jps
root@ingesta:~# ls -la /home/iabd_admin/flume_ingest
total 20
drwxr-xr-x 1 root root 4096 Nov  5 17:56 .
drwxr-xr-x 1 root root 4096 Nov  5 17:56 ..
drwxr-xr-x 2 root root 4096 Nov  5 17:56 .flumespool
-rw-r--r-- 1 root root   60 Nov  5 17:53 log_file_01.txt.COMPLETED
root@ingesta:~# hdfs dfs -ls /user/iabd_admin/ingesta/weblogs
Found 1 items
-rw-r--r--   3 root supergroup         60 2026-11-05 17:57 /user/iabd_admin/ingesta/weblo
root@ingesta:~#
```

Prueba de B3

```
root@ingesta:~# ls -ld /home/iabd_admin/flume_ingest
drwxr-xr-x 1 root root 4096 Nov  5 18:04 /home/iabd_admin/flume_ingest
root@ingesta:~# grep spoolDir ~/flume_configs/weblogs.conf
agent.sources.s1.spoolDir = /home/iabd_admin/flume_ingest
```

Prueba de B4

```
Info: Including Hive libraries found via () for Hive access
+ exec /usr/lib/jvm/java-8-openjdk-amd64//bin/java -Xmx20m -Xmx512m -Dflume.root.logger=INFO,console -cp '/opt/flume/conf:/opt/flume/lib/*:/etc/hadoop:/opt/hadoop-3.2.1/share/hadoop/common/lib/*:/opt/hadoop-3.2.1/share/hadoop/common/*:/opt/hadoop-3.2.1/share/hadoop/hdfs:/opt/hadoop-3.2.1/share/hadoop/hdfs/lib/*:/opt/hadoop-3.2.1/share/hadoop/hdfs/*:/opt/hadoop-3.2.1/share/hadoop/mapreduce/lib/*:/opt/hadoop-3.2.1/share/hadoop/mapreduce/*:/opt/hadoop-3.2.1/share/hadoop/yarn:/opt/hadoop-3.2.1/share/hadoop/yarn/lib/*:/opt/hadoop-3.2.1/share/hadoop/yarn/*:/lib/*' -Djava.library.path=:/opt/hadoop-3.2.1/lib/native org.apache.flume.node.Application --conf-file /home/iabd_admin/flume_configs/weblogs.conf --name agent
SLF4J: Class path contains multiple SLF4J bindings.
SLF4J: Found binding in [jar:file:/opt/flume/lib/log4j-slf4j-impl-2.18.0.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: Found binding in [jar:file:/opt/hadoop-3.2.1/share/hadoop/common/lib/slf4j-log4j12-1.7.25.jar!/org/slf4j/impl/StaticLoggerBinder.class]
SLF4J: See http://www.slf4j.org/codes.html#multiple_bindings for an explanation.
SLF4J: Actual binding is of type [org.apache.logging.slf4j.Log4jLoggerFactory]
```

```
192.168.1.5 - POST /login.php
10.0.0.7 - GET /productos.html
10.0.0.9 - POST /compra.php
root@ingesta:~# cleaqr
bash: cleaqr: command not found
root@ingesta:~# clear
root@ingesta:~# ls -ld /home/iabd_admin/flume_ingest
drwxr-xr-x 1 root root 4096 Nov  5 18:04 /home/iabd_admin/flume_ingest
root@ingesta:~# grep spoolDir ~/flume_configs/weblogs.conf
agent.sources.s1.spoolDir = /home/iabd_admin/flume_ingest
root@ingesta:~# jps
1009 Application
1134 Jps
root@ingesta:~#
```

```
root@ingesta:~# ls -la /home/iabd_admin/flume_ingest
total 24
drwxr-xr-x 1 root root 4096 Nov  5 18:20 .
drwxr-xr-x 1 root root 4096 Nov  5 18:04 ..
drwxr-xr-x 2 root root 4096 Nov  5 18:04 .flumespool
-rw-r--r-- 1 root root   60 Nov  5 17:53 log_file_01.txt.COMPLETED
-rw-r--r-- 1 root root   59 Nov  5 18:04 log_file_02.txt.COMPLETED
root@ingesta:~# hdfs dfs -ls /user/iabd_admin/ingesta/weblogs
Found 2 items
-rw-r--r--   3 root supergroup         60 2026-11-05 17:57 /user/iabd_admin/ingesta/weblogs/FlumeData.1793901379645
-rw-r--r--   3 root supergroup         59 2026-11-05 18:05 /user/iabd_admin/ingesta/weblogs/FlumeData.1793901869522
root@ingesta:~#
```

```
root@ingesta:~# echo "192.168.1.30 - GET /tienda.html" > ~/log_temp.txt
root@ingesta:~# echo "192.168.1.31 - POST /pago.php" >> ~/log_temp.txt
root@ingesta:~# mv ~/log_temp.txt /home/iabd_admin/flume_ingest/log_file_03.txt
root@ingesta:~# sleep 5

root@ingesta:~# ls -la /home/iabd_admin/flume_ingest
total 28
drwxr-xr-x 1 root root 4096 Nov  5 18:23 .
drwxr-xr-x 1 root root 4096 Nov  5 18:23 ..
drwxr-xr-x 2 root root 4096 Nov  5 18:23 .flumespool
-rw-r--r-- 1 root root   60 Nov  5 17:53 log_file_01.txt.COMPLETED
-rw-r--r-- 1 root root   59 Nov  5 18:04 log_file_02.txt.COMPLETED
-rw-r--r-- 1 root root   62 Nov  5 18:23 log_file_03.txt.COMPLETED
root@ingesta:~#
```

```
root@ingesta:~# hdfs dfs -ls /user/iabd_admin/ingesta/weblogs
Found 3 items
-rw-r--r--   3 root supergroup         60 2026-11-05 17:57 /user/iabd_admin/i
ngesta/weblogs/FlumeData.1793901379645
-rw-r--r--   3 root supergroup         59 2026-11-05 18:05 /user/iabd_admin/i
ngesta/weblogs/FlumeData.1793901869522
-rw-r--r--   3 root supergroup         62 2026-11-05 18:24 /user/iabd_admin/i
ngesta/weblogs/FlumeData.1793903019779
root@ingesta:~#
```

Discusión y cierre

1. ¿Por qué no usamos Sqoop para los logs? ¿Qué pasaría?

Sqoop no resulta adecuado para esta tarea porque requiere una fuente JDBC con tablas estructuradas (--connect jdbc:... y --table), mientras que los logs son simples ficheros de texto plano sin esquema. Además, se trata de una herramienta orientada a ejecuciones puntuales mediante trabajos MapReduce —lo que introduce una latencia incompatible con el envío «al instante» que requiere el escenario—, de modo que para capturar nueva información habría que relanzarlo continuamente. Por último, al carecer el archivo de texto de una columna de control como una fecha o un ID secuencial, Sqoop no puede realizar cargas incrementales (--incremental) y terminaría releyendo y copiando todo el fichero en cada ejecución.

2. ¿Por qué no usamos Flume para la base de datos? ¿Se podría?

Sqoop no es la opción adecuada aquí porque depende del conocimiento previo del esquema, leyendo los metadatos de la tabla para generar la clase correspondiente (como ocurrió al consultar las columnas de pedidos), además de ofrecer consistencia al copiar la tabla completa de una sola vez. Por contra, un origen que consulta periódicamente corre el riesgo de duplicar o perder filas, carece del paralelismo por claves que permite a Sqoop dividir la carga entre mappers, y presenta problemas de fiabilidad, ya que el uso de un memory channel implica que cualquier caída del sistema causaría la pérdida de las filas ya leídas.

3. ¿Qué riesgo tiene el memory channel en producción?

El principal problema de este enfoque radica en el elevado riesgo de pérdida de datos y falta de persistencia, ya que los eventos en memoria desaparecen por completo si el agente se reinicia, se agota la memoria (OutOfMemoryError) o se produce una caída del equipo. Esta vulnerabilidad se acentúa por la ventana de riesgo que genera el proceso de lectura: el

source marca un archivo como .COMPLETED en cuanto introduce los eventos en el channel, en lugar de esperar a su llegada efectiva a HDFS.